

Gas Absorption in a Packed Bed

Group #3

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Introduction

- **Purpose of lab**
 - To determine the fluid flow characteristic in a packed bed gas absorber
 - including the behavior of the flooding point
 - To determine the behavior of gas absorption and calculation of the overall mass transfer coefficient
 - when CO₂ gas is removed from a stream of air, using water as solvent

Theory

- **Calculation of Void Volume**

$$\epsilon = \frac{V_{void}}{V_{apparent}} \text{ or } \epsilon = 1 - \frac{\rho_B}{\rho_s} \quad (1)$$

Where:

ϵ = void volume

V_{void} = volume of the voids (cm³ or mL)

$V_{apparent}$ = total volume of the packed bed (cm³ or mL)

ρ_s = density of solid packing (g/mL)

ρ_B = bulk density of the packed particles (g/mL)

Theory continued...

- When the packing is dry, and the liquid flow rate is zero ($L=0$) the relationship between mass velocity and pressure drop per unit height is linear

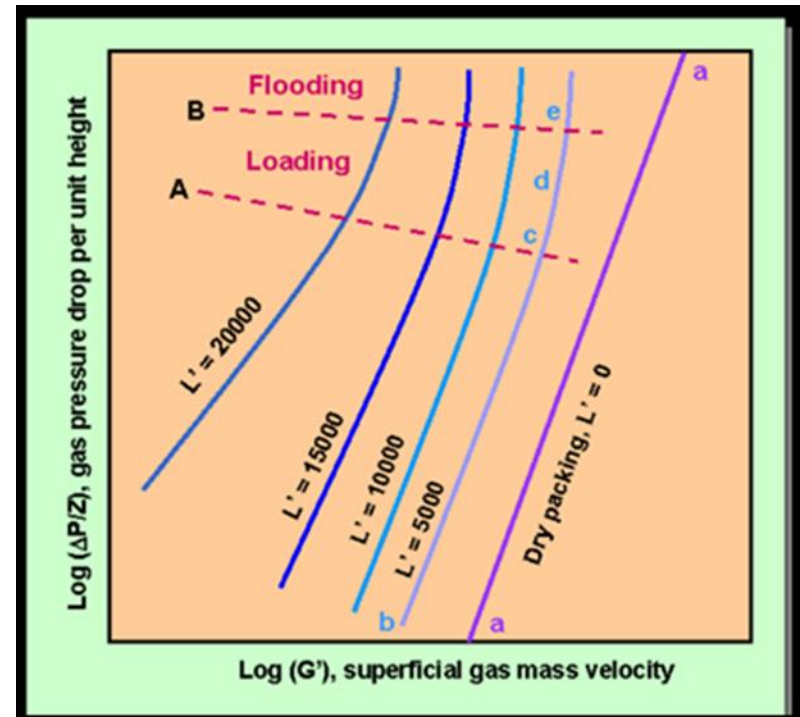


Figure 1: Fluid Characteristic in a Packed Bed Gas Absorber [3]

Theory continued...

- When the liquid flow is kept constant and the gas velocity is increased the relationship between the superficial gas velocity and the gas pressure drop per unit height can be observed in Figure 1, line b

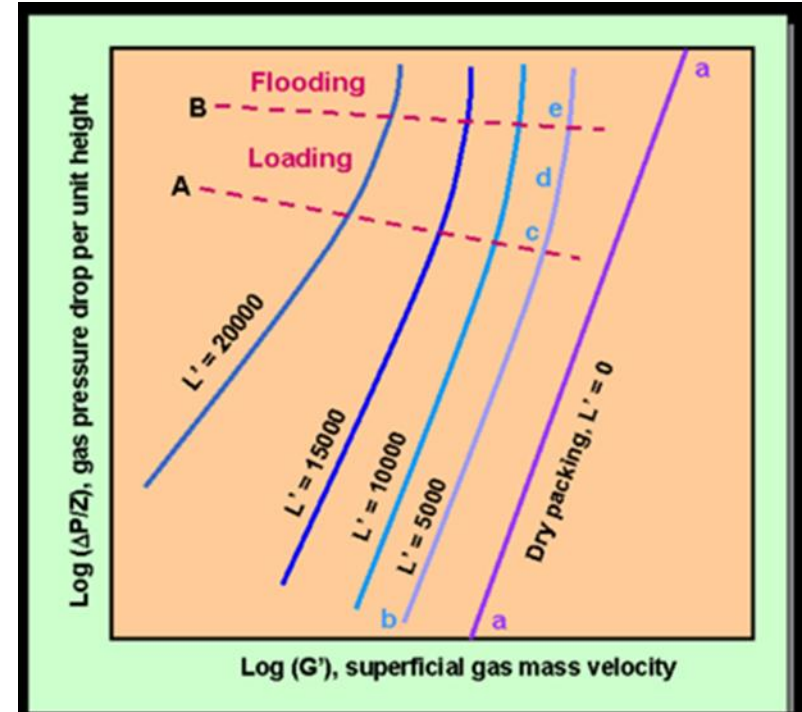


Figure 1: Fluid Characteristic in a Packed Bed Gas Absorber [3]

Theory continued...

- At low to moderate gas velocities the behavior of the relationship is linear.
- As the velocity of the gas is increased the pressure drop observed will also increase.
- At greater pressure drops a change in slope occurs and can be observed as point C in Figure 1, also known as the loading point

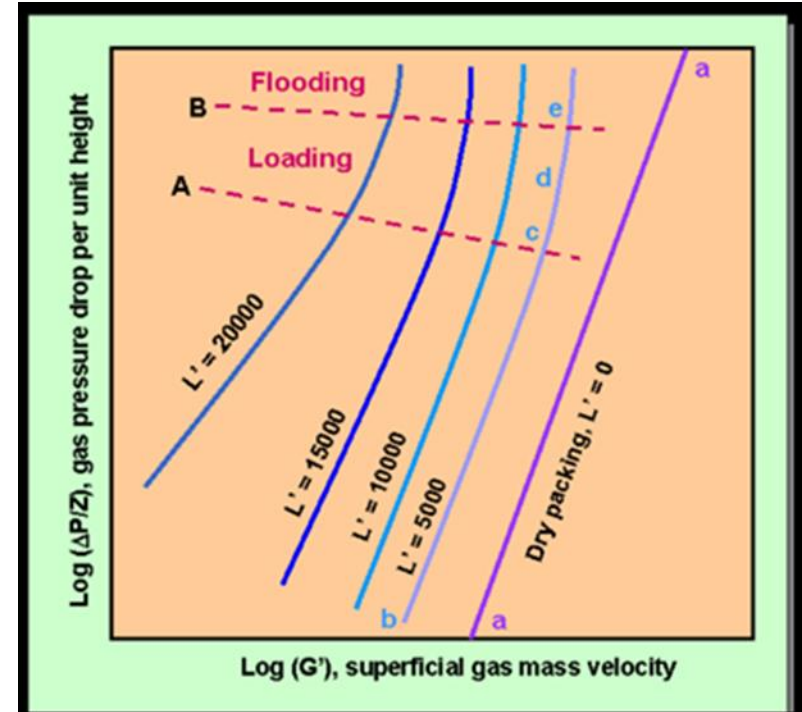


Figure 1: Fluid Characteristic in a Packed Bed Gas Absorber [3]

Theory continued...

- Flooding point occurs at point e
- Flooding occurs when the gas flow rate is much greater than the liquid flowrate.
- Gas can push through the column and lift out of the packed bed decreasing the contact between the gas and liquid phase and decreases the effectiveness of the absorption

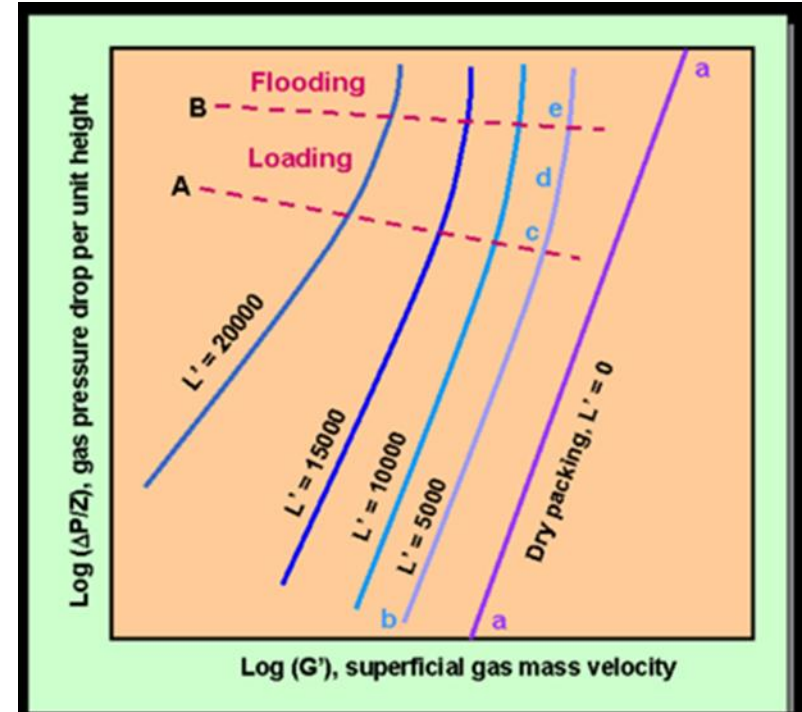


Figure 1: Fluid Characteristic in a Packed Bed Gas Absorber [3]

Theory continued...

- Gas Absorption Characteristics in a Packed Bed Gas Absorber**

$$\frac{\text{Rate of mass transfer}}{\text{Volume}} = K_y a (y_A^* - y_A) \quad (2)$$

$$\frac{\text{Rate of mass transfer}}{\text{Volume}} = K_x a (x_A - x_A^*) \quad (3)$$

Where:

K_y = mass-transfer coefficient for vapor

K_x = mass-transfer coefficient for liquid

a = interface area per unit volume of the column (m^2/m^3)

*typical units for $K_y a$ and $K_x a$ are $\frac{\text{kmol}}{(\text{s} \cdot \text{m}^3)}$

x_A = bulk liquid mole fraction

y_A = bulk vapor mole fraction

y_A^* = vapor mole fraction in equilibrium with bulk liquid mole fraction (x_A)

x_A^* = liquid mole fraction in equilibrium with bulk vapor mole fraction (y_A)

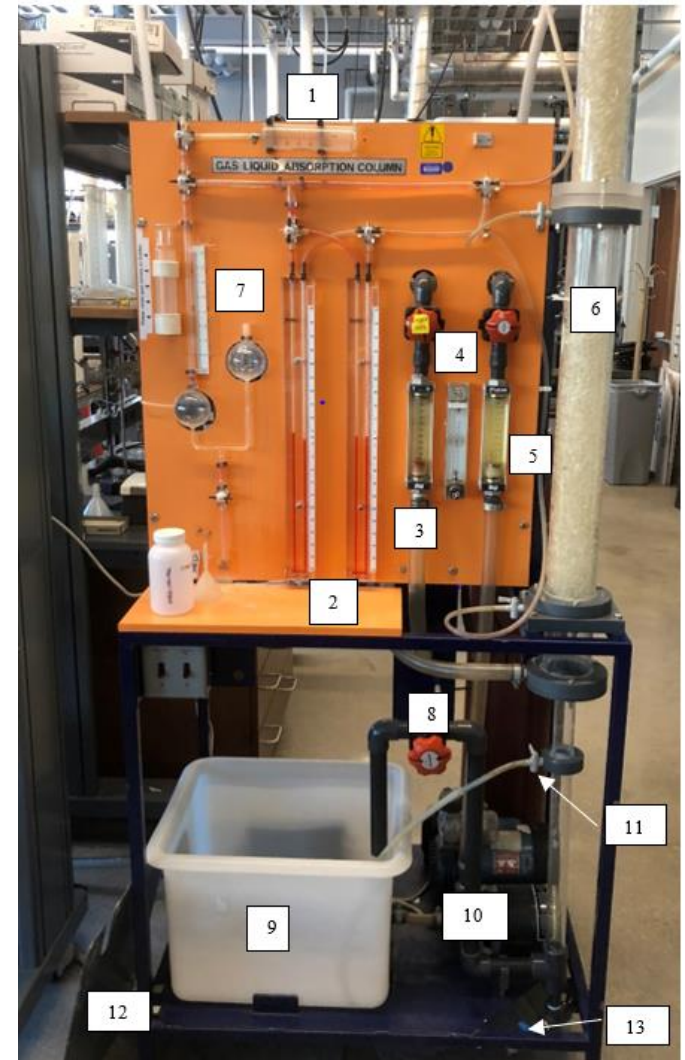
Theory continued...

- The $(y_A^* - y_A)$ and $(x_A - x_A^*)$ terms are also known as the driving force and can also be partial pressure and concentration in addition to mole fractions. The concentration can be calculated below:

- $$C_d \left(\frac{\text{gram mole}}{\text{liter of free CO}_2} \right) = \left(\frac{V_B * \text{Conc. Base}}{\text{mL of Sample}} \right) \quad (4)$$

Apparatus

- 1 – Piston
- 2 – Manometer
- 3 – Air flow meter
- 4 – CO₂ flow meter
- 5 – Water flow meter
- 6 – Packed column
- 7 – Hempl apparatus
- 8 – Control valve
- 9 – Fluid reservoir
- 10 – Centrifugal pump
- 11 – Sample draining valve
- 12 – Tank draining valve
- 13 – Column draining valve
- 14 – Water supply tube



Material & Supplies

Materials	Description
Compressed air	Provides air flow into the gas stream
Phenolphthalein indicator solution	Used in titration of sample from Hempl apparatus
0.05 N or 0.01N NaOH solution	Standard solution used in titration procedure
CO ₂	Carbon dioxide gas in gas stream
Deionized water	Liquid in reservoir tank; Wets the piston of the syringe to prevent gas from leaking out to the atmosphere
1.0 N NaOH solution	Liquid contained inside the absorption globes in Hemple apparatus
50 mL burette	Holds standard NaOH solution for titration
Burette clamp	Holds burette in place
50 mL graduated cylinder	Contains collected samples directly from the liquid outflow point S4
Stopwatch	Time air flow through the column and time intervals when expelling air from the pump
125 mL Erlenmeyer flasks	Contains liquid samples for titration procedure
Thermometer	Measures temperature of water in reservoir tank
Safety gloves	Protect extremities from NaOH
Goggles	Protect the eyes from chemical splashes
Parafilm	Seals flasks to prevent evaporation of liquid samples

Procedure

Part 1

Void Fraction

- Fill graduated with Raschig rings
- Top up with water and determine amount required

Column Flooding

- Start with 0 L/min of water, and 0 L/min CO₂
- Increase air flow rate accordingly
- Noticeable with higher flow rates of water
- Take note of flow rate settings when flooding occurs

Procedure

Part 2a

Absorption of CO₂ by H₂O in Equilibrium Operation

1. Collect initial sample of CO₂ and find concentration via titration
2. Run column with CO₂ and water
 1. Run airflow at 20 L/min
 2. Run waterflow at 6 L/min
 3. Run CO₂ at the assigned flowrate
3. Sample CO₂ concentration after 10 minutes of run time
4. Continue sampling until CO₂ values are constant

Procedure

Part 2b

Titration

- Ensure water level at the drainage valve is sufficiently high for obtaining samples
- Draw at desired time interval (1-2 minutes)
- Titrate with 0.5N NaOH until endpoint is achieved

Hemph Column

- Use piston and syringe to flush the lines releasing into the atmosphere
- Isolate absorption globe and close vent to atmosphere then use the syringe to draw in the CO₂
- Vent syringe to atmosphere and close after 10 seconds and reconnect to absorption globe
- Close piston into the absorption globe and draw piston out again
- Note level in the indicator tube

Procedure

Part 3

Absorption of CO₂ by H₂O in continuous column mode

1. Collect initial sample and measure the concentration of CO₂
2. Run column continuously
 1. Adjust water flowrate to 1.5 L/min
 2. Adjust air flowrate to 20 L/min
 3. Adjust CO₂ flow to assigned flowrate
3. Sample:
 1. 30 second intervals for 2 minutes
 2. 1-minute intervals for 5 minutes
 3. 5-minute intervals until steady state
4. Analyze gas fraction using Hempl apparatus

Experimental Challenges

- Make sure to have titration section organized and ready to go. If possible, have two to three group members titrating to help manage time.
- Seal the conical flask with parafilm to prevent the CO₂ from escaping.
- Pay attention to water level in the column above sample drainage valve.

Safety

- Wear safety goggles
- Gloves – especially important for titration
- MSDS for CO₂ tank and Sodium Hydroxide
- Dispose of titrated solutions into waste bottle located in the fume hood